Loss Functions as Noise Models

Signal Detection and Estimation Theory, Project 3

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The problem

- Each loss function is the negative log-likelihood of one noise model.
- MSE assumes Gaussian noise. MAE assumes Laplace noise.
- **Question:** what is the cost when the noise model is wrong?

Model. We estimate a constant A from N samples:

$$x[n] = A + w[n], \qquad n = 0, \ldots, N-1, \qquad w[n] \text{ i.i.d., zero mean.}$$

This is linear regression $\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \mathbf{w}$ with $\mathbf{X} = \mathbf{1}$.

Measure of cost.

$$\text{efficiency ratio} = \frac{\text{Var}(\hat{A})}{\text{CRLB}}$$

A ratio of 2 means twice the variance of the best estimator. If both variances fall as 1/N, the estimator needs 2 times more data. If the variance stops falling, more data will not help.

The loss comes from the noise model

Gaussian noise

$$\ln p(\mathbf{x}; A) = c - \frac{1}{2\sigma^2} \sum_n (x[n] - A)^2$$

The MLE minimises the squared error.

$$\Rightarrow \hat{A} = \bar{x} \quad \text{(mean)}$$

Laplace noise

$$\ln p(\mathbf{x}; A) = c - \frac{1}{b} \sum_n |x[n] - A|$$

The MLE minimises the absolute error.

$$\Rightarrow \hat{A} = \text{median}(\mathbf{x})$$

- For the median, the derivative of $\sum |x - A|$ is (samples below A) $-$ (samples above A). It changes sign at the middle sample.
- Choosing a loss is thus to choose a noise model!

CRLB and the predicted cost

CRLB

- Gaussian: σ^2/N
- Laplace: b^2/N

Variance of each estimator

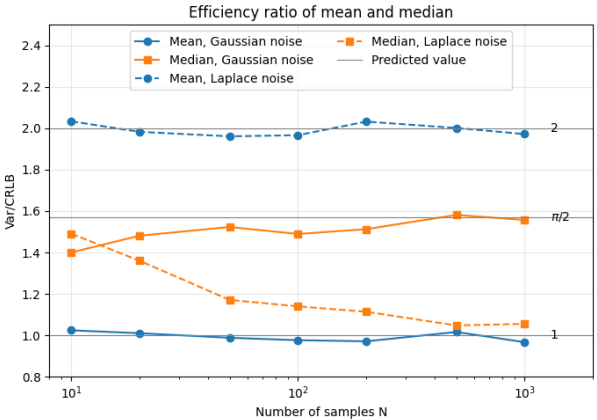
- Mean: $\mathrm{Var}(w)/N$ (exact for all N)
- Median: $\approx 1/(4N\,p_w(0)^2)$ (large N)

Predicted efficiency ratio

	Gaussian	Laplace
Mean (MSE)	1	2
Median (MAE)	$\pi/2$	1

- Each estimator is best only for its own noise.
- Under Laplace noise, the mean needs 2 times more data than the median.

Simulation with Gaussian and Laplace noise

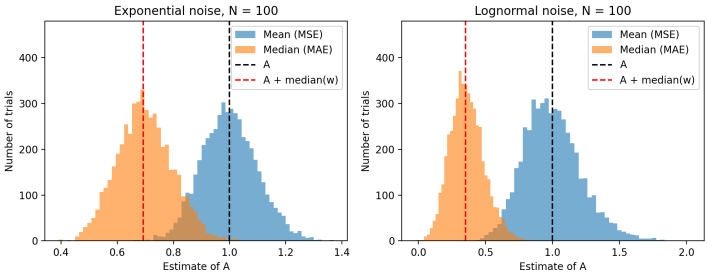


5000 trials. Measured at $N = 100$:

	Gauss.	Laplace
Mean	0.97	2.01
Median	1.51	1.17

- The mean is flat for all N.
- The median reaches its value only for large N.
- At small N, our median results match exact published values (NIST, 1964).

Skewed noise: the loss changes the target



Consider the case of zero-mean noise, but now not symmetric!

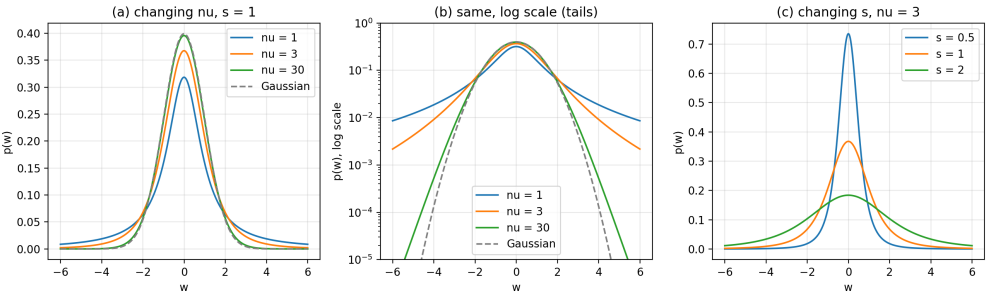
- Mean $= A$
- Median $= A + \text{median}(w)$

	Pred.	Meas.
Expon.	0.693	0.698
Lognorm.	0.351	0.363

MSE is not biased. MSE and MAE estimate different values.

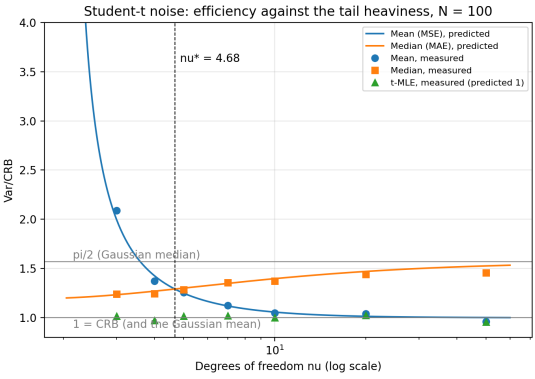
Student-t noise: the model

$$p_w(w) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)s}\left(1+\frac{w^2}{\nu s^2}\right)^{-\frac{\nu+1}{2}}$$



- ν sets the tails. $\nu \to \infty$ gives Gaussian. $\nu = 1$ gives Cauchy.
- s sets the width. $\mathrm{Var}(w) = s^2\nu/(\nu-2)$, finite only for $\nu > 2$.
- It is a Gaussian with a random variance (Lange et al., 1989).

Student-t noise: results



$$\mathrm{CRLB} = \frac{\nu+3}{\nu+1} \cdot \frac{s^2}{N}$$

$$\frac{\mathrm{Var}(\bar{x})}{\mathrm{CRLB}} = \frac{\nu(\nu+1)}{(\nu-2)(\nu+3)}$$

- Heavy tails make the mean much worse (leftwards in the plot).
- The median stays between 1.2 and 1.5.
- They are equal at $\nu \approx 4.68$.
- The t-MLE is a weighted mean with weight $(\nu+1)/(\nu+r^2)$. It reaches the CRLB.

Thank you. Questions?